Ernesto Tellez Perez

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EDUCATION

University of California, Berkeley | GPA: 3.9 — **Berkeley, CA**
Bachelor of Science, Mechanical Engineering — *Graduating December 2027*
Regent's and Chancellor's Scholar
Relevant coursework: Solid Mechanics, Fluids, Dynamics, Composites, Manufacturing and Design

Sierra College | GPA: 4.0 — **Rocklin, CA**
Associate of Science for Transfer, Mathematics and Physics — *Graduated August 2025*

EXPERIENCE

Stanford University — Biomimetics and Dexterous Manipulation Lab — **Stanford, CA**
Summer Undergraduate Research Fellow (SURF) — *June 2026 - Present*

- Designed and built a compact FTIR system for the lab's NASA TechLeap proposal, capturing a 3 × 3 inches field of view with the camera within 2 in. of the glass.
- Designed and built an 8 × 8 in. FTIR test bench to measure gecko-inspired adhesive contact during engagement and failure while minimizing gravity-assisted loading.
- Used Python, OpenCV, and ChArUco calibration to correct lens distortion, scale images, and measure individual contact patches.

Formula Electric at Berkeley — **Berkeley, CA**
Aerodynamics and Structural Engineer — *October 2025 - Present*

- Led the design and ANSYS structural analysis of the front-wing mounting system, reducing its mass by 33% while maintaining a minimum factor of safety of 1.75.
- Converted aerodynamic data into load cases and boundary conditions for front-wing FEA.
- Reduced the total front-wing mass by 15% through structural redesign and composite manufacturing.
- Manufactured carbon-fiber aerodynamic components using layup preparation and resin infusion, while supporting composite fabrication for other subteams.

MathWorks — **Remote**
Project Intern — *July 2025 - August 2025*

- Analyzed shear and moment forces in a drone structure using MATLAB to ensure stability and integrity within a 4 lb maximum payload constraint.
- Optimized the center of mass and thrust alignment to maximize payload capacity and improve flight performance using limited 1 lb thrust motors.
- Led team design meetings, delegated tasks, and reviewed each member's work to ensure accuracy and progress.
- Compiled findings into a final report for presentation to MathWorks judges.

PROJECTS

Plug-&-Forget Smart Health Toilet Sensor
Design Team — *July 2025 - August 2025*

- Developed a functional, clip-on smart toilet sensor for daily monitoring of renal and metabolic health through pH levels and urine color analysis.
- Prototyped 2 sensors: color sensor, analyzed RGB values of urine color ranked for hydration levels, and pH sensor, analyzed acidity and basicity of urine indicating concerns with renal & metabolic health.
- Designed the housing in CAD, then utilized 3D printing and laser cutting through multiple iterations.
- Presented a prototype to 100+ industry and academic professionals at PREP/TPREP Design Expo.
- Awarded People's Choice and Best Prototype.

SKILLS

Software: Ansys Structural, SolidWorks, Fusion 360, AutoCAD, OnShape, MATLAB, Python, Java, Photoshop
Technical: Finite Element Analysis, Topology, Design for Manufacturing, Waterjet, 3D Printing, Laser Cutting, Project Management, Photography